2023 IEEE International Conference on High Performance Computing & Communications, Data Science & Systems, Smart City & Dependability in Sensor, Cloud & Big Data Systems & Application (HPCC/DSS/SmartCity/DependSys)

A Privacy-Preserving Timing Attacks Mitigation in Information-Centric Edge Networks

Ningchun Liu*,†, Shuai Gao†, Teng Liang‡, Xindi Hou†, Lei Yu§, Hongke Zhang†

*Research Institute of Intelligent Networks, Zhejiang Lab, Hangzhou 311121, China
†School of Electronic and Information Engineering, Beijing Jiaotong University, Beijing 100044, China
‡Department of New Networks, Peng Cheng Laboratory, Shenzhen 518066, China
§Intelligent Cloud Network Operation Center, China Telecom Zhejiang Co.,Ltd, Hangzhou 310009, China
Email: nchliu@zhejianglab.edu.cn; {shgao, freezerburn, hkzhang}@bjtu.edu.cn;
liangt@pcl.ac.cn; yul34@chinatelecom.cn

***Abstract*—In-network caching is one of the most important features of Information-Centric Networking (ICN), which reduces data-retrieval latency, especially at edge networks. However, in-network caching can lead to privacy leakage with *timing attacks*, i.e., an adversary retrieves cached data and identifies it based on the difference of round trip time (rtt) of cached and non-cached data retrieval. Existing mitigation mechanisms either abandon in-network caching or intentionally add latency to cached data retrieval but fails to prove their effectiveness. In this paper, we propose a timing attack mitigation mechanism based on differential privacy exponential mechanism that strictly protects content privacy against timing attacks without significantly increasing the content retrieval delay. Theoretical analysis and evaluation results show that our mechanism protects content privacy effectively at a low cost.**

***Index Terms*—Information-centric edge networks, timing attacks, differential privacy.**

I. Introduction

Information-Centric Networking (ICN) is considered a promising architecture for edge networks due to its architectural benefits, such as in-network caching, mobility support, and off-the-grid communication support [1]–[4]. Specifically, in-network caching can pervasively cache frequently accessed data at edge routers, which reduces data retrieval latency, improves network resource utilization, and alleviate the workload on the server side. However, in-network caching poses risks of content privacy leakage, i.e, an adversary is able to identify and retrieve the content recently accessed by a consumer according to the in-network caching on an edge router with the method of *timing attacks*, which counts on the round-trip time (rtt) difference between the cached data and non-cached data.

Fig. 1 illustrates the timing attacks with an example. An adversary can determine whether a consumer within the same edge router has accessed a specific Youtube video by sending requests (i.e., *Interest* packets) with the name prefix of both the Youtube video and another Netflix video. The edge router will cache the Youtube video previously accessed by the consumer, making it possible for the request to be fulfilled locally. In contrast, the Netflix video must be retrieved from a remote server. By analyzing the difference in RTT between the two requests, the consumer's privacy can be compromised.

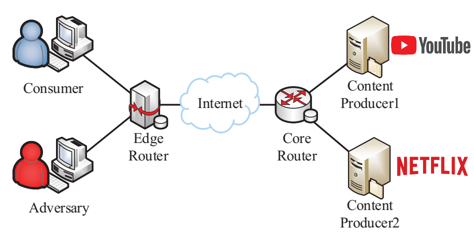

Fig. 1. Typical example of timing attacks.

Several mechanisms have been proposed to deal with timing attacks, which can be broadly classified into three categories, namely (i)detection mechanisms for timing attacks [5]–[7], (ii)cache policy-based mitigation mechanisms [8]–[11], and (iii)delay-based mitigation mechanisms [12]–[14]. Detection mechanisms focus on attack detection and lack the consideration of attack mitigation. In the cache-based mechanisms, the router no longer caches all content copies, which violates the in-network caching feature of ICN. In addition, the change of caching strategy brings the cost to all routers in the network and is difficult to deploy. Delay-based mechanisms introduce random delays. However, random delays with a certain distribution are difficult to strictly protect content privacy, and also adds cost to the consumer side.

Based on the above observation, the previous work is faced with the following three open issues, namely (i) strict privacy protection: mitigation mechanism should strictly protect content privacy, and that privacy should be given a formal description, and the attacker cannot determine the privacy of cache based on the collected RTT sample information, (ii) low cost: mitigation mechanism should not significantly increase the average delay of consumer content retrieval, although this is a tradeoff. Also, the cache hit rate of network nodes should not be significantly reduced, and (iii) easy deployment: mitigation mechanism should be easy to deploy. On the one hand

2023 IEEE International Conference on High Performance Computing & Communications, Data Science & Systems, Smart City & Dependability in Sensor, Cloud & Big Data Systems & Application (HPCC/DSS/SmartCity/DependSys) | 979-8-3503-3001-4/23/$31.00 ©2023 IEEE | DOI: 10.1109/HPCC-DSS-SMARTCITY-DEPENDSYS60770.2023.00013

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DOI 10.1109/HPCC-DSS-SmartCity-DependSys60770.2023.00013

changes to the network should be kept to a minimum to avoid adding network entities. On the other hand, modifications to the network nodes should be lightweight.

This paper proposes a strict privacy-preserving and lightweight mechanism based on differential privacy (DP) to mitigate timing attacks. The main contributions are summarized as follows.

- We propose a DP-based timing attack mitigation mechanism, which can protect content privacy without significantly affecting the consumers' content retrieval delay.
- We design a DP-based delay generation method. Based on this method, we propose a privacy-preserving in-network caching response algorithm to add noise to the potentially malicious content request RTT.
- We conduct simulations to evaluate the proposed mechanism. The simulation results show that our mechanism can mitigate timing attacks effectively at a low cost.

II. Related Work

A. Cache policy-based Mitigation Mechanisms

Abani *et al.* [8] proposes a caching strategy mechanism based on betweenness centrality in the intermediate routers. Sensitive content is cached in routers with higher betweenness centrality to increase the number of consumers using that router, thereby increasing the anonymity set size of consumers and protecting content privacy. Sivaraman *et al.* [10] introduces a privacy-enhanced caching strategy mechanism that first formulates a privacy-enhanced distributed optimization problem and converts it into a non-cooperative game problem involving multiple consumers, aiming to protect consumer privacy at a minimal cost. Jones *et al.* [9] proposes a collaborative caching mechanism, where routers are divided into several groups, and each group of routers collaborates to cache content in one router in the group, thereby protecting content privacy. These approaches all change the caching strategy to prevent all routers from caching the same content replica, thus protecting content privacy, which actually violates the caching characteristic of ICN. Furthermore, cache policy-based solutions consume a significant amount of router computing power, reduce network performance, and require significant network changes, making them challenging to deploy.

B. Delay-based Mitigation Mechanisms

Mohaisen *et al.* [12] introduces a delay-based mechanism for mitigating timing attacks in which a specific delay is added to the content request response cached by the router, thus protecting content privacy. This delay simulates the delay generated when content is provided by one of the upstream routers or servers. Acs *et al.* [13] proposes a threshold-based mechanism for mitigating timing attacks, which introduces cache misses at the router and continues forwarding the content request to the upstream router when the number of content requests is less than a random threshold. Kumar *et al.* [14] designs a cache privacy detection method in which detection technology is deployed on the router and delay is only introduced when a timing attack is detected. While delay and cache misses can ensure the privacy of requested content in the aforementioned solutions, they significantly reduce the efficiency of caching because there are content cache replicas along the way, but consumers still face significant delays.

III. Attack Model

We describe the attack model in Fig. 2. First, the consumer sends an Interest packet with the prefix "/bjtu/file1", and the corresponding Data packet will be cached in the edge router alongside the forwarding path. To determine whether the consumer has accessed the content with the prefix "/youtube/moive1", the adversary sends Interest packets with the prefix "/youtube/moive1" and "/youtube/moive2" successively. Then, the adversary records the RTT when receiving the corresponding Data packets. By comparing the difference in RTTs, the adversary can spy on the consumer's privacy, that is, whether the consumer has accessed the content prefixed with "/youtube/moive1".

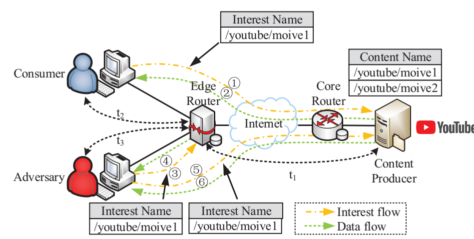


Fig. 2. Attack model of timing attacks in information-centric edge networks.

Assuming that $RTT_1 = t_3$ is the RTT from the ICN edge router, $RTT_2 = t_1 + t_3$ is the RTT from the content producer, RTT_e is the RTT of the adversary requesting content, and ε is a negligible time difference. After collecting the above time samples, the adversary will make a decision based on the following conditions.

Case 1: $|RTT_e - RTT_1| < \varepsilon$.

Decision 1: Content is cached in the edge router.

Case 2: $|RTT_e - RTT_2| < \varepsilon$.

Decision 2: Content is only in the content producer.

Case 3: $RTT_e > RTT_1$, $RTT_e < RTT_2$ and $|RTT_2 - RTT_1| \gg \varepsilon$.

Decision 3: Content is cached in routers in the forwarding path.

IV. Design of Mitigation Mechanism

In this section, we first introduce the concept of differential privacy, especially the exponential mechanism. Then, we present the DP-based delay generation method and show the design details of our timing attacks mitigation mechanism. Finally, we conducted a theoretical analysis of how our mechanism protects privacy.

A. Preliminaries

Differential privacy is a privacy-preserving framework proposed by Dwork [15], which makes it impossible for an adversary to infer any user's data record from the noise-perturbed data. The definition of differential privacy is as follows.

Definition 1: ε-differential privacy. A random algorithm $\mathcal{M}$ with domain $\mathbb{N}^{|\mathcal{X}|}$, where χ is all A collection of entries, $\mathbb{N}^{|\mathcal{X}|}$ is the space formed by all possible sub-databases.If for all $\mathcal{S} \subseteq Range(\mathcal{M})$ and for all adjacent databases $x, y \in \mathbb{N}^{|\mathcal{X}|}$, $\|x-y\|_1 \leq 1$, satisfy $\Pr[\mathcal{M}(x) \in \mathcal{S}] \leq \exp(\varepsilon)\Pr[\mathcal{M}(y) \in \mathcal{S}] + \delta$. Then $\mathcal{M}$ satisfies (ε, δ)-differential privacy. If $\delta = 0$, then $\mathcal{M}$ satisfies ε-differential privacy.

Definition 2: Sensitivity. Given a utility function u : $\mathbb{N}^{|\mathcal{X}|} \times \mathcal{R} \rightarrow \mathbb{R}$, where $\mathbb{N}^{|\mathcal{X}|}$ is the quotation database, $\mathcal{R}$ is certain pricing, and $\mathbb{R}$ is the pricing generated on the quotation database, whose sensitivity is defined as

$$\Delta u = \max_{r \in \mathcal{R}, \{x,y \mid \|x-y\|_1 \leq 1\}} |u(x,r) - u(y,r)|$$

where $u(x,r)$ is a utility function and x, y represent two adjacent databases with only one element difference. Thus, sensitivity refers to the maximum difference in the output of $u(x,r)$ in any two adjacent databases.

Definition 3: Exponential Mechanism. The differential privacy exponential mechanism means that $\mathcal{M}(x,u,\mathcal{R})$ outputs an element $r \in \mathcal{R}$ with the probability proportional to $\exp\left(\frac{\varepsilon u(x,r)}{2\Delta u}\right)$. Further, $\exp\left(\frac{\varepsilon u(x,r)}{2\Delta u}\right)$ needs to be normalized to the probability value $\Pr\left[\mathcal{M}(x,u,\mathcal{R}) = r\right] = \frac{\exp\left(\frac{\varepsilon u(x,r)}{2\Delta u}\right)}{\sum\limits_{r' \in \mathcal{R}} \exp\left(\frac{\varepsilon u(x,r')}{2\Delta u}\right)}$.

B. Privacy Definition

Assuming that C is the content that needs to be privacy-preserving, t_C represents the RTT value of the requested content C, and $t_{\mathcal{R}}$ refers to the RTT of another content request. A random algorithm $\mathcal{M}$ with domain $\mathbb{N}^{|\mathcal{X}|}$, for all $\mathcal{R} \subseteq Range(\mathcal{M})$ and for all adjacent databases $x, y \in \mathbb{N}^{|\mathcal{X}|}$, $\|x-y\|_1 \leq 1$ satisfies $\Pr[t_C \in \mathcal{R}] \leq \exp(\varepsilon)\Pr[t_{\mathcal{R}} \in \mathcal{R}]$. According to Definition 1, $\mathcal{M}$ satisfies ε-differential privacy. In other words, since the delay set $\mathcal{R}$ satisfies ε-differential privacy, the privacy of content C is protected, and the strength of privacy protection is ε. The privacy of the content C is that the attacker cannot control the RTT value t_C of the requested content C in the delay set $\mathcal{R}$ It can be effectively discriminated against the RTT value $t_{\mathcal{R}}$ of requesting other content.

C. DP-based Delay Generation

First, we construct the set $\mathcal{R} = \left\{r = i * \frac{t_x}{h}\right\}(i = 0, 1, \ldots, h-1, h)$, where h represents the hop between the edge router and the content producer, and t_x refers to the RTT for the edge router requesting content from the content producer. Then, we select and output a content request RTT value with probability $\Pr\left[\mathcal{M}(x,u,\mathcal{R}) = r\right] = \frac{\exp\left(\frac{\varepsilon u}{2\Delta u}\right)}{\sum\limits_{r \in \mathcal{R}} \exp\left(\frac{\varepsilon u}{2\Delta u}\right)}$, which satisfies Definition 3, where ε represents the privacy budget, and Δu represents the sensitivity. To reduce the cost caused by privacy protection, the utility function u should be a decreasing function, that is, the element with a larger value in the set $\mathcal{R}$ should be selected with a smaller probability. We choose the exponential mechanism instead of the Laplace mechanism because of the irreversibility of time.

D. Design of Timing Attacks Mitigation Mechanism

We present the design details of our mechanism in this section, the core of which is the privacy-preserving cache response algorithm, which can obfuscate the original distribution of consumer content request RTT, thereby strictly protecting content privacy. Meanwhile, this mitigation mechanism is only deployed in edge routers.

Algorithm 1 The privacy-preserving cache response algorithm

Input:
Interest, $Int = (f, n, t_0)$, where f refers to the incoming interface, and n represents the name prefix, t_0 represents the timestamp;
Output:
Respond to Interest with a privacy-preserving delay;
Initialization: Let $\chi(f,n)$ = 0, where $\chi(f,n)$ means the number of times n is requested under the interface f;
Process Interest: Obtain the timestamp t_0, record the current time t_1, and calculate $t = t_1 - t_0$;
if $\chi(f,n) == 0$ **then**
  if n is cached **then**
    Delay t_d;
    Forward Data packet to f;
  else
    Send Interest with name prefix n, record the arrival time of the Data packet t_2, and calculate $t_x = t_2 - t_1$ //RTT from edge router to content producer;
    Calculate $h = \frac{t_x}{2t} + 1$ //average hops from consumer to content producer;
    Generate t_d; //in Section IV-C;
    $\chi(f,n)++$;
    Forward Data packet to f;
  end if
else
  Forward cached Data packet to f;
end if

In the initialization, $\chi(f,n)$ represents the times of requesting content n in the interface f. When the edge router receives the Interest packet, it first obtains the incoming interface f, the content name prefix n, and the timestamp t_0. Then, record the current time t_1, and calculate the one-way delay $t = t_1 - t_0$ from the consumer to the edge router. Next, make a judgment, if $\chi(f,n) = 0$, then further judge whether the content n has been cached. If n is cached, it means that consumers in other interfaces of the edge router have requested the content n, then after waiting for a delay of t_d, forward the corresponding Data packet to interface f. If n is not cached, then the edge router will send an Interest packet with the prefix name n to request the content n from the content producer. After receiving the reply Data packet, record its arrival time t_2, and calculate the RTT $t_x = t_2 - t_1$ from the edge router to the content producer and the average hops $h = \frac{t_x}{2t} + 1$, where the average number of hops h is the parameter for generating the waiting delay t_d. Then, update $\chi(f, n)$, and forward the Data packet to interface f. In addition, if $\chi(f, n) \neq 0$, it means that users under the interface f have requested the same content n multiple times. According to the in-network cache feature of ICN, the cached Data packet will be forwarded directly to the interface f.

E. Privacy Analysis

We theoretically prove that our mechanism maintains content privacy in this section. Consider the outputs $x \in t_{\mathcal{C}}$ and $y \in t_{\mathcal{R}}$ of two adjacent databases.

$$
\begin{aligned}
\frac{\Pr\left[\mathcal{M}(x, t_C) \in \mathcal{R}\right]}{\Pr\left[\mathcal{M}(y, t_{\mathcal{R}}) \in \mathcal{R}\right]} &= \frac{\left(\frac{\exp\left(\frac{\varepsilon u(x,r)}{2\Delta u}\right)}{\sum\limits_{r' \in \mathcal{R}} \exp\left(\frac{\varepsilon u(x,r')}{2\Delta u}\right)}\right)}{\left(\frac{\exp\left(\frac{\varepsilon u(y,r)}{2\Delta u}\right)}{\sum\limits_{r' \in \mathcal{R}} \exp\left(\frac{\varepsilon u(y,r')}{2\Delta u}\right)}\right)} \\
&= \left(\frac{\exp\left(\frac{\varepsilon u(x,r)}{2\Delta u}\right)}{\exp\left(\frac{\varepsilon u(y,r)}{2\Delta u}\right)}\right) \cdot \left(\frac{\sum\limits_{r' \in \mathcal{R}} \exp\left(\frac{\varepsilon u\left(y,r'\right)}{2\Delta u}\right)}{\sum\limits_{r' \in \mathcal{R}} \exp\left(\frac{\varepsilon u(x,r')}{2\Delta u}\right)}\right) \\
&= \exp\left(\frac{\varepsilon\left(u\left(x,r\right) - u\left(y,r\right)\right)}{2\Delta u}\right) \cdot \left(\frac{\sum\limits_{r' \in \mathcal{R}} \exp\left(\frac{\varepsilon u\left(y,r'\right)}{2\Delta u}\right)}{\sum\limits_{r' \in \mathcal{R}} \exp\left(\frac{\varepsilon u(x,r')}{2\Delta u}\right)}\right) \\
&\leq \exp\left(\frac{\varepsilon}{2}\right) \cdot \exp\left(\frac{\varepsilon}{2}\right) \cdot \left(\frac{\sum\limits_{r' \in \mathcal{R}} \exp\left(\frac{\varepsilon u\left(x,r'\right)}{2\Delta u}\right)}{\sum\limits_{r' \in \mathcal{R}} \exp\left(\frac{\varepsilon u(x,r')}{2\Delta u}\right)}\right) \\
&= \exp\left(\varepsilon\right)
\end{aligned}
$$

It is proved that $\mathcal{M}(x, t_C)$ can obtain ε-differential privacy guarantee.

V. Performance Evaluation

In this section, combined with the differential privacy library Diffprivlib [16] released by IBM, we implement the proposed differential privacy-based timing attack mitigation mechanism on the ndnSIM. In addition, we evaluate the performance from two metrics: (1) RTT of content retrieval and (2) cache hit ratio of edge routers.

A. Experiment Environment

We construct a typical tree-shaped network topology on the ndnSIMv2.5 [17], as shown in Fig. 3. Specifically, the topology contains two content producers, sixteen forwarding nodes, an attacker, and sixteen legitimate consumers represented by "leaf-n". The forwarding nodes consist of eight core routers and eight edge routers denoted by "bb-n" and "cgw-n", respectively. Table. I provides a summary of the link parameters in the topology. Note that our mechanism is deployed on edge routers. In addition, the ndnSIMv2.5 runs in a laptop with a 2.20GHz*4 Intel Core i7-3632QM processor, 8GB memory, and Ubuntu 16.04 operating system.

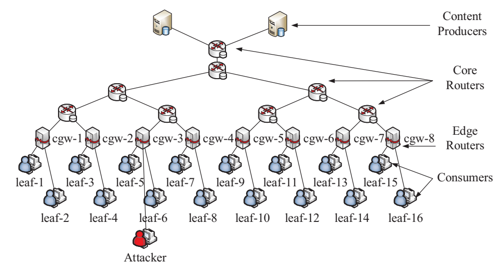


Fig. 3. Experiment environment.

TABLE I
EXPERIMENTAL PARAMETER.

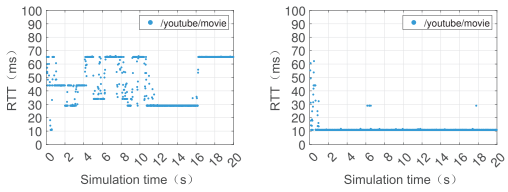
Id	Description	Parameter	
		Delay	Bandwith
1	Content producer-Core router	4.77ms	1000Mbps
2	Core router-Core router	[2.51,7.56]ms	1000Mbps
3	Edge router-Core router	[3.11,9.10]ms	100Mbps
4	Consumer-Edge router	5ms	10Mbps

In the simulation, "leaf-5" and "leaf-6" send fifty Interest packets per second successively and uniformly with the name prefix "/youtube/movie". Other legitimate consumers send 50 Interest packets uniformly with random name prefixes "/youtube/movie" or "/netflix/movie". The attacker sends 100 Interest packets per second uniformly, of which 50 Interest packets with the name prefix "/netflix/movie" to obtain the RTT from himself to the remote content producer and other 50 Interest packets with the name prefix "/youtube/movie" to detect whether consumers located in the same edge router have accessed that content.

B. RTT of Content Retrieval

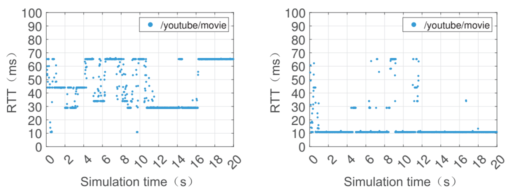
To verify the effectiveness of our mechanism, we analyze the RTT of content retrieval in three scenarios: without timing attacks (in Fig. 4), with timing attacks (in Fig. 5), and with timing attack mitigation (in Fig. 6). The RTT of content retrieval refers to the time interval between the consumer (or the attacker) sending the first Interest packet and receiving the last corresponding Data packet. Furthermore, to evaluate the cost of our mechanism, a comparative evaluation is conducted with the representative Mohaisen mechanism [12] (in Fig. 7), which is a typical delay-based timing attack mitigation mechanism.

In Fig. 4, the RTT of the "leaf-6" is stable, while the "leaf-5" fluctuates up and down. This is because the in-network caching feature of ICN makes the "leaf-6" obtain the requested content from the edge router, while the "leaf-5" need pull the content from the content producer.

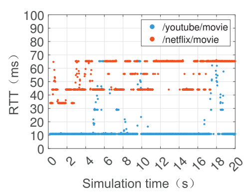


(a) Leaf-5. (b) Leaf-6.

Fig. 4. RTT of the consumer without timing attacks.

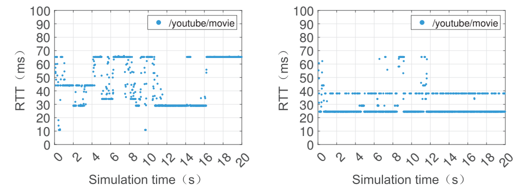


(a) Leaf-5. (b) Leaf-6.

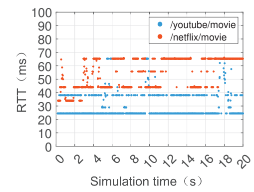


(c) Attacker.

Fig. 5. RTT of consumer and attacker with timing attacks.

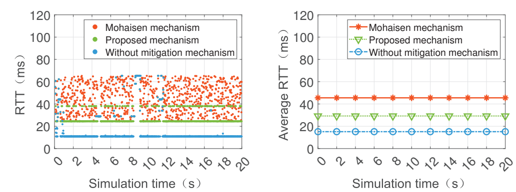


(a) Leaf-5. (b) Leaf-6.



(c) Attacker.

Fig. 6. RTT of consumer and attacker with timing attacks mitigation.



(a) Original. (b) Average.

Fig. 7. RTT of the consumer in different mechanisms.

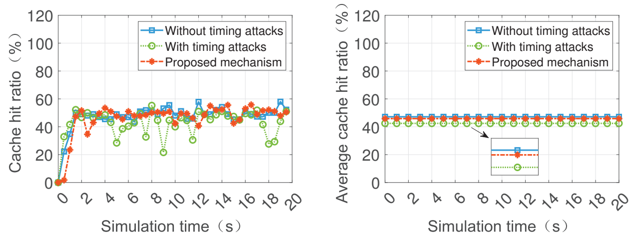
As shown in Fig. 5(a) and Fig. 5(b), the RTT of leaf-5 and leaf-6 with timing attacks is the same as that without timing attacks, which shows that timing attacks are concealed and not easy to be perceived by consumers. In Fig. 5(c), the RTT of content with the name prefix "/youtube/movie" is mostly equal to 10.9 ms, which indicates that "leaf-5" or "leaf-6" in the same edge router requested the same content previously, and the content is satisfied by the edge router. In addition, the RTT of content with the name prefix "/netflix/movie" is discretized and in the range of 30 ms to 70 ms, which means that this content is not cached in the edge router. Therefore, the privacy of "leaf-5" and "leaf-6" is compromised.

In Fig. 6(a), the RTT of "leaf-5" has a slight change, which is due to the timing attack mitigation mechanism only dealing with the repeated content requests. Fig. 6(b) shows that the content request RTT of the "leaf-6" has changed significantly with timing attacks , and mostly equal to 24.5 ms. This is because our mechanism adds the DP-based delay to the content request RTT. Fig. 6(c) indicates that the content request RTT with the name prefix "/youtube/movie" and "/netflix/movie" is discretized. The attacker can only infer that the content with the name prefix "/youtube/movie" is located in the routers closer to itself, which indicates that our mechanism can protect content privacy.

As shown in Fig. 7, we takes "leaf-6" as an example to show the RTT of consumer content requests in different mechanisms. In Fig. 7(a), the original values show that the RTT of content request in the proposed mechanism is mostly in the range of 20 ms to 40 ms, mainly 24.5 ms. In the Mohaisen mechanism, the content request RTT is in the range of 24.5 ms to 65.3 ms with a uniform distribution. This is because, unlike the Mohaisen mechanism in which the time delays are randomly generated, the DP-based delay generation method allows smaller elements in the delay set to be selected with higher probability by setting the utility function u reasonably, while strictly protecting privacy. Fig. 7(b) shows the average RTT of consumers in different mechanisms. The average RTT of the Mohaisen mechanism and our mechanism is 45.2 ms and 29.4 ms respectively, which means that our mechanism has 35% lower cost while strictly preserving privacy compared with the Mohaisen mechanism. Our mechanism can further reduce the cost by adjusting the privacy budget according to different privacy protection strengths.

C. Cache Hit Ratio of Edge Routers

To evaluate the impact of the proposed mechanism on the resource utilization in the network, the cache hit rate of the edge router nodes is selected as a parameter and compared in three scenarios: without timing attacks, with timing attacks, and with timing attack mitigation.



(a) Original.

(b) Average.

Fig. 8. Cache hit ratio of the edge router.

Fig. 8 shows the cache hit rate of edge router ”cgw-3” as an example. The raw data in Fig. 8(a) shows that the cache hit rate of the edge router fluctuates significantly when the timing attack occurs compared to when the timing attack does not occur. This is due to the attacker requesting content with the name prefix ”/netflix/movie” that was not previously cached by the edge router, thus generating cache contention. With the proposed timing attack mitigation mechanism enabled, the cache hit rate of the edge router tends to smooth out because the proposed mechanism adds noise satisfying the differential privacy index mechanism to each cache hit request, and this noise objectively makes the time interval between cache hit requests processed by the edge gateway longer, reducing the probability of cache contention. Fig. 8(b) shows the average value of node cache hit rate. On the one hand, the timing attack leads to a decrease in the average cache hit rate of the edge gateway, which corroborates the conclusion of the original data. Also, our mechanism improves the resource utilization of the edge networks.

VI. Conclusion

In this paper, we have proposed a DP-based timing attack mitigation mechanism, which can protect content privacy without significantly affecting the consumers’ content retrieval delay. We have also designed a DP-based delay generation method, Based on this, a privacy-preserving in-network caching response algorithm has been proposed. Several simulations have been conducted to evaluate our mechanism and compared it with other representative mechanisms. Simulation results show that our mechanism is effective to mitigate timing attacks at a low cost. In the future, we will validate our mechanism with more realistic topologies in an actual system.

Acknowledgment

This work was supported by the National Key Research and Development Project of China (No. 2022YFB2901503), the National Natural Science Foundation of China (No. U22A2005 and No. 61972026), Key Research Project of Zhejiang Lab (No. 2021LE0AC02).

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